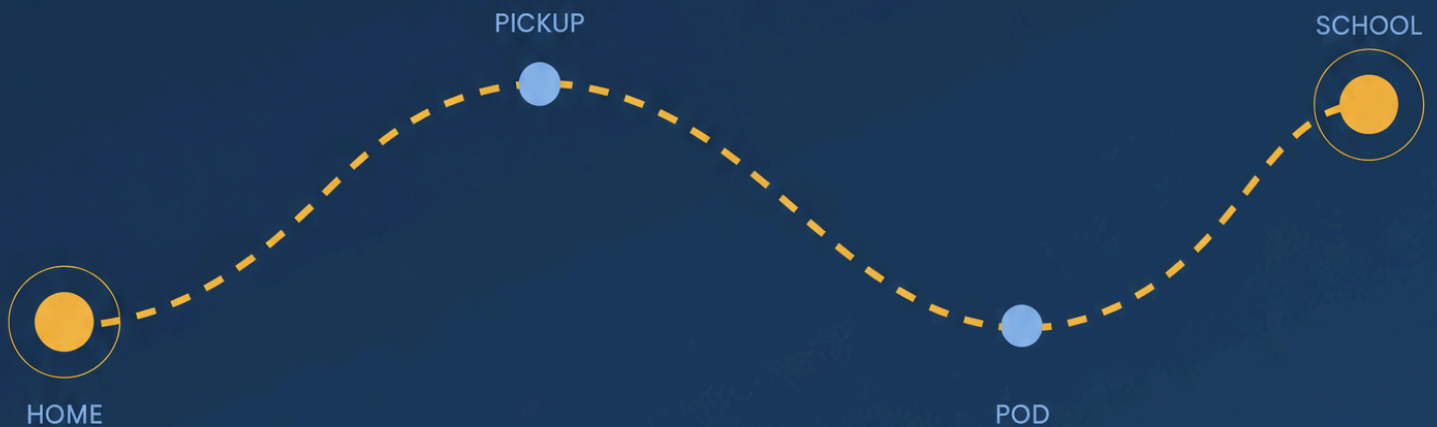


Sarathi

A supervised mobility network for India's youngest commuters

From CabDriver to CareDriver



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How to read this document The sections marked **Assumption** are working hypotheses to validate. The sections marked **Target** are future goals. The sections marked **USP** are the distinctive product choices retained from the original concept.

Step 0 — Comprehend the situation

0.1 Situation in one minute

Sarathi is a scheduled, parent-paid school commute for Classes 1–5. It uses a familiar crew, school-supported handoffs, mixed vehicle types, and a custody log. The first launch is one dense tier-1-city pocket; the assumptions are tested through field interviews and a live route pilot.

0.2 Questions I will ask before building

These are the six questions that should be answered through interviews, field visits, and live tests.

No.	Question	What I need to learn
1	Who pays, who uses, and who controls dismissal?	Parent, child, school, and guardian responsibilities.
2	Which trips and timings are truly repeated?	Morning and afternoon routes, school calendars, and staggered timings.
3	Should Sarathi own vehicles or onboard operators?	Driver supply, vehicle quality, insurance, and control requirements.
4	Will families accept a scheduled subscription?	Demand for fixed routes versus on-demand flexibility.
5	What price and vehicle mix can work?	Willingness to pay across auto, van, and cab options.
6	What transport expense already exists?	Current bus, van, auto, car, and informal monthly spend.

0.3 How big is the market?

This is an illustrative planning funnel, not a validated market size. It narrows from a working urban-population base to children aged 6–11, private-school enrolment, households able to pay, and families already paying for school travel. Replace every layer with enrolment data, route-level supply data, and a parent payment survey before launch.

	Stage	Meaning
120M	Population	Eight tier-1 metros
9.6M	Age group	Children aged 6–11
4.8M	School segment	Fee-paying private schools
2.4M	Affordability	Household can pay around Rs 4,000/month
1.0M	Immediate segment	Already paying for transport

Illustrative market-sizing funnel; every number is an assumption to validate.

0.4 Why the ride-hailing model changes

For a child, the product must replace dynamic matching with a familiar crew, post-ride rating with continuous checks, and a map-only experience with a custody record. The child is not expected to verify the driver or call support independently.

0.5 US benchmark versus Indian launch context

The US examples are useful benchmarks, but Sarathi should not copy their buyer, fleet, or pricing model. Zum’s public model is district-led and technology-heavy: it connects students, drivers, and school districts through real-time operations and reports serving more than 5,000 schools and districts, with public performance claims and 68 million rides in 2025. HopSkipDrive is also primarily institution-led: it partners with school districts and provides supplemental transportation through vetted CareDrivers for cases such as foster care, special-needs, and students not served by regular bus routes. [[Zum official source](#)] [[HopSkipDrive official source](#)]

Dimension	US benchmark pattern	Why India needs a different starting point
Buyer	Districts, public agencies, schools, and institutional programmes.	Start parent-paid but school-supported; the school adds trust and access without becoming the sole buyer.
Supply	Large school-bus operations plus vetted CareDrivers or contracted fleets.	Use existing autos, vans, cabs, and small buses; recruit by route, vehicle type, and school-time availability.
Primary need	Supplement traditional routes, cover special programmes, and improve district efficiency.	Solve the everyday home-school commute where informal transport is familiar but custody is weak.
Trust mechanism	District contracts, regulated operators, driver vetting, and connected dashboards.	Familiar locked crew, school countersignature, authorised collector, and a visible handoff ledger.
Technology posture	Connected mobility platforms and large operating systems.	Begin with a small custody log, ETA rules, and local AI support; add hardware only after the manual process proves the need.
Strategic choice	Scale institutional coverage and fleet efficiency.	Choose India because dense routes, mixed supply, existing family transport spend, and school-led trust make a focused subscription pilot testable without owning a fleet.

The Indian choice is therefore not “US model versus no model.” It is **US operating principles adapted to Indian constraints**: reliable handoffs, vetted adults, proactive communication, and school coordination, delivered through a more flexible local supply base.

Step 1 — Build user personas

The four primary perspectives

User	Job to be done	Design implication
Parent	I want proof that my child reached school safely without repeatedly calling anyone.	Give a simple status, a reliable timeline, and reason-coded exceptions.
Child	I want the commute to feel normal and not make me feel watched or singled out.	Use a familiar crew, a discreet band, and a human attendant.
Driver / attendant	I need predictable work, clear routes, and simple safety steps I can complete every trip.	Give a reliable schedule, fair vehicle-specific pay, checklists, escalation support, and a portable safety record.
School	I need to know which child left with which adult, on record.	Introduce the dismissal console and independent handoff signature after the parent, child, and crew workflows are understood.

The supply-side users

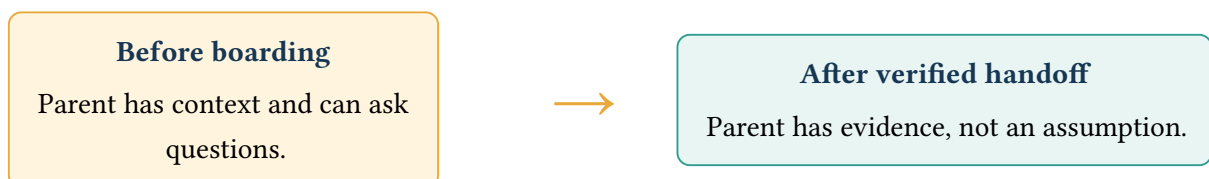
Ramesh, the driver, needs predictable income, full seats, and no cash collection. Sunita, the attendant, needs respectable work that fits a split shift and gives her a path to grow. Their incentives are not secondary: without reliable crews, the parent-facing product cannot operate.

Research improvement The persona stories should be labelled as either interview evidence or working assumptions. Add the number and type of participants before treating these details as validated.

Step 2 — Map out the journey

Figure 2 — The dark information window

The commute is more than a period of travel. For the parent, it contains a window in which useful information can disappear. Anxiety rises when the child boards and falls when arrival and handoff are confirmed.



The parent's information gap is the product opportunity.

The "dark window" is a useful hypothesis to test with parent diary studies. The product should not encourage compulsive checking; it should provide a small number of meaningful milestones.

Figure 3 — Four custody transfers

The full trip contains four custody points: home parent to crew, crew to school gate staff, school gate staff to the returning crew, and crew to the authorised home collector. The returning crew is represented by the same driver-and-attendant node on the return leg. Each point should have a system record plus an independent human confirmation where practical, with a manual escalation path if a record is missing.



A custody log makes “safe” observable and auditable.

Journey framework: Lewis C. Lin’s 5Es The journey is checked through **Entice** (the trigger), **Enter** (booking and first steps), **Engage** (the commute and handoffs), **Exit** (verified home handoff), and **Extend** (renewal, feedback, and exception learning). This is a brainstorming framework, not evidence about demand. [5Es source]

Step 3 — Understand user pain points

The pain points are grouped into six areas: parent before the ride, parent during the ride, parent around the ride, child comfort and fear, vehicle capacity, and driver operations.

Area	Observed or assumed pain	Product response
Before ride	No clear record of driver, vehicle, insurance, or recourse.	Repeated vetting, insurance proof, and a formal contract.
During ride	No trustworthy answer when the vehicle is late or deviates.	Calm tracking, reason codes, and route corridor lock.
Around ride	No backup when the driver is ill; unclear collectors and billing.	Backup crew, authorised collectors, absence credits, and autopay.
Child	Heat, fear, discomfort, boredom, and tracker stigma.	Attendant, familiar crew, discreet band, and no entertainment tablets in v1.
Capacity	Overloading can create a short-term incentive for an informal operator; validate this through ride-alongs rather than treating it as a fact.	Manifest and headcount reconciliation with a hard capacity cap.
Driver	Cash chasing and no portable proof of safe work.	Central collection, guaranteed payout, and a verified crew record.

Figure 4 — Frequency versus severity

The prioritisation matrix separates frequent and severe pains from rare but catastrophic ones. The latter still deserve engineering because their downside is unacceptable.



Ignore in v1

Low-impact convenience issues

P1 quick wins

Absence · billing · lost items

Prioritisation should consider both frequency and severity.

Step 4 — The other user: the driver

The driver and attendant are not just supply. They are users whose schedule, income, vehicle, and judgement determine whether the safety promise works.

4.1 Start from the work that already exists

The initial model assumes a mix of auto, van, and cab drivers. They may have different vehicle capacities, licence requirements, routes, and preferred price bands. Sarathi should therefore pay by vehicle type, route distance, operating hours, and role—not force every driver into one price.

The school commute should be a predictable block of work rather than a full-day employment change. A driver can retain other work outside the agreed school windows, while Sarathi centralises collection, route allocation, and safety records.

4.2 A smaller, more trustworthy supply estimate

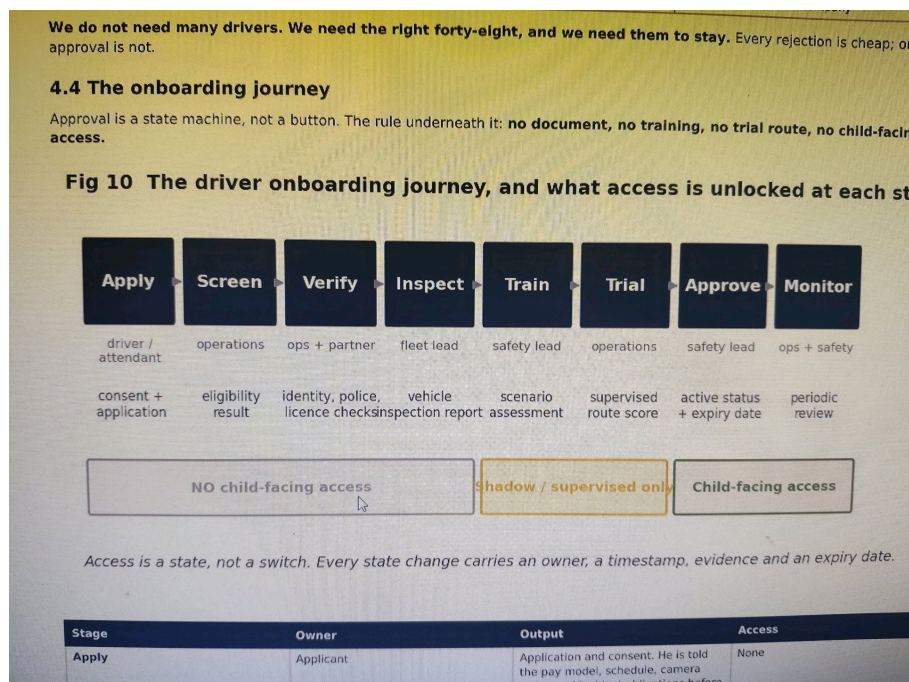
The broad metro driver pool is large, but only a fraction is relevant to school transport. This is a range-based assumption, not a claim that one final number can serve every vehicle type.

Stage	Directional range	Interpretation
Commercial drivers in one metro	3–5 lakh	Broad pool across autos, vans, cabs, and small buses.
Suitable school-time availability	10–20%	Drivers willing to work fixed morning and afternoon windows.
Suitable vehicle and route fit	10–25%	Vehicle size, school access, capacity, and corridor match.
Interested after pay and safety are explained	20–35%	Interest depends on payout, empty seats, and operating friction.
Clear baseline checks and supervised trial	60–75% of applicants	A reasonable quality gate without pretending the pass rate is 5%.
First-pilot requirement	100+ active crews	Enough variety for three schools and multiple vehicle types; refine through field sourcing.

The practical conclusion is: **start with the number of routes the first three schools can actually fill, then recruit by vehicle type and corridor.**

4.3 The onboarding journey

Approval is a state machine, not a button, but the first test should be understandable and proportionate: Apply → Screen → Verify → Inspect and train → Supervised trial → Active and monitored.



Reduced reference view of the driver-onboarding journey.

No child-facing access is granted before identity, licence, vehicle, safeguarding, and supervised-route checks are complete. A failed safety or safeguarding check stops the application; a minor documentation issue can be corrected and rechecked.

4.4 Supply-side USP

A vetted driver receives a portable record of training, checks, and incident history. That record becomes a recruiting and retention benefit, while the parent gets a more accountable counterparty than an informal cash-only arrangement.

4.5 A proportionate driver test

The test should be demanding enough to protect children but not so narrow that it rejects every viable auto, van, and cab operator. The initial cohort can be evaluated as follows:

Stage	Illustrative outcome per 100 applicants	What is checked
Screened	100	Availability, licence category, vehicle type, route fit, and pay expectations.
Baseline checks clear	60–70	Identity, licence, documents, references, and police/antecedent check where required.
Vehicle and training clear	45–55	Vehicle condition, capacity, safeguarding, handoffs, emergency response, and privacy.

Supervised trial clear	30–40	Driving behaviour, punctuality, calm communication, headcount discipline, and one exception scenario.
Active pilot crew	25–35	Final safety-lead approval with expiry date and periodic recheck.

This range is more credible than a single 5% pass-rate claim. It still rejects unsafe applicants, but it leaves room for different driver categories and improves as sourcing becomes more targeted.

Step 5 — Cut through prioritisation

The prioritisation method is a RICE-style adaptation with Impact measured as **harm avoided**, not revenue generated. This is a project-specific adaptation, not a universal industry standard.

Feature	Priority	Reason
Locked crew and pod	P0	Familiarity reduces unknown-adult risk.
Headcount and capacity interlock	P0	Prevents overloading and the child-left-behind failure.
Handoff log	P0	Makes custody auditable across parent, school, and crew.
Route corridor lock	P0	Makes deviations visible without crying wolf on every traffic diversion.
Calm live tracking	P0	Answers the parent's binary question without compulsive checking.
Silent SOS band	P1	Important but low-confidence until tested with children.
Parent-viewable cabin video	Never in v1	Creates privacy and consent problems for other children.
In-ride tablets	Never in v1	Adds supervision, theft, and motion-sickness burden.

MVP boundary

The smallest credible pilot should include one school pocket, one vehicle tier, manual crew vetting, a fixed route, a manifest, QR or RFID handoffs, basic parent notifications, and a staffed escalation number. The pilot should prove safe repeatability before adding face matching, distress-audio models, and a hardware shutdown interlock.

The advanced hardware remains a strong USP, but it should be introduced only after the manual process identifies which steps truly need automation.

Step 6 — The solution

Three ways to commute

Tier	Vehicle	Illustrative hard cap	Crew	Illustrative price	Restriction
Sarathi Auto	3-wheeler	3 children	Verified driver	Rs 2,400/month	Class 3–5, short corridor
Sarathi Van	13-seater	12 children	Locked driver + attendant	Rs 4,000/month	Full product; core v1
Sarathi Cab	Sedan	3 children	Driver; attendant on request	Rs 6,500/month	Long routes, Class 1–2, special protocol

The hard capacity rule is the central operating promise. The capacities and price bands below are illustrative pilot assumptions, to be checked against vehicle permits, local rules, route economics, and payment tests. The manifest, band count, and occupancy count must agree. If the vehicle is over capacity or the final cabin count is not zero, the trip cannot start or close.

6.2 One school day, end to end

The figure below keeps the original product logic. Green is the happy path, blue is human judgement, and red is a hard stop. The flow begins with a recurring school trip and ends only after a verified home handoff and an empty-cabin check.

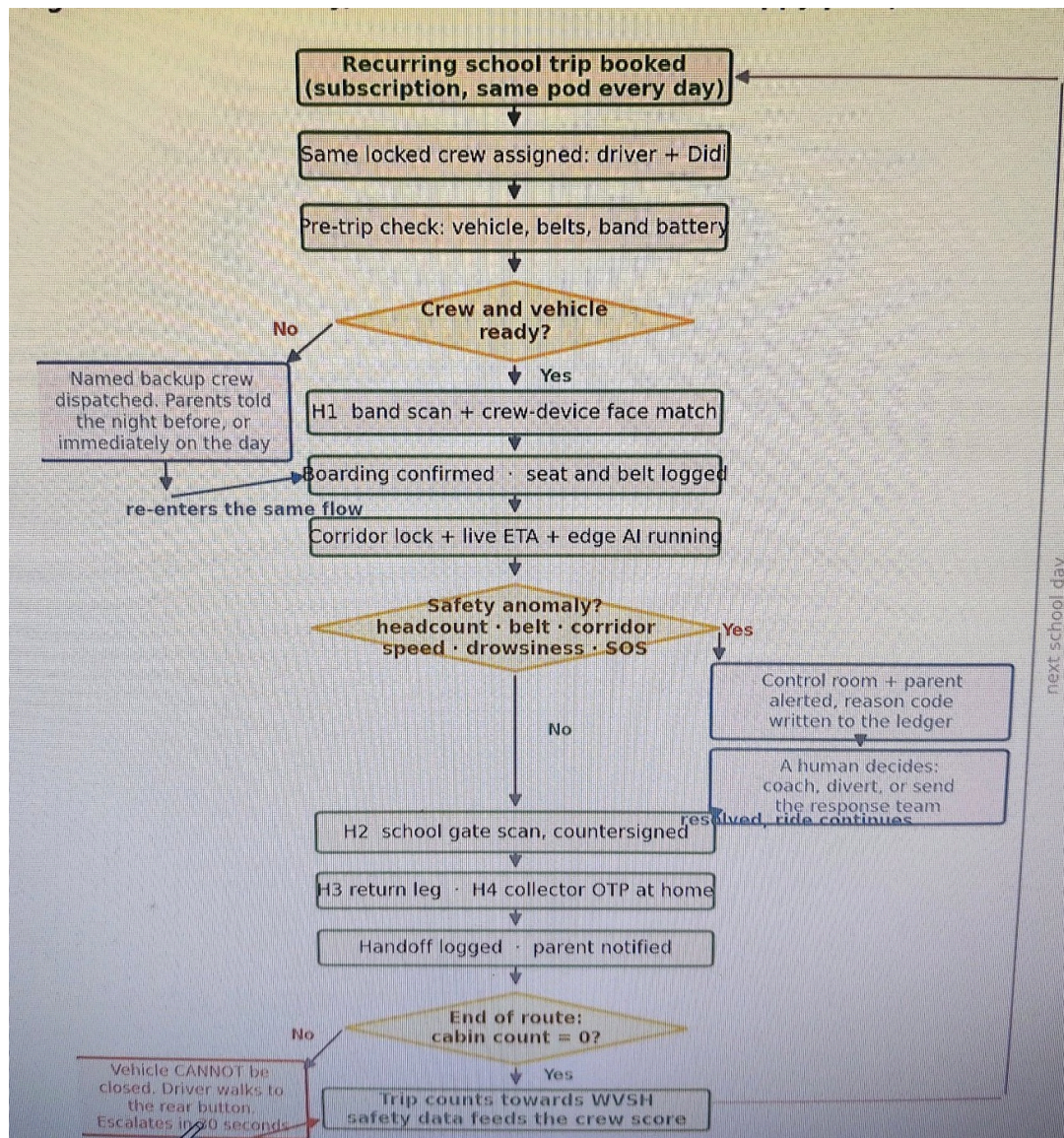


Fig. 13 — One school day, end to end. Green is the happy path; blue is human judgement; red is the hard stop. The important product decision is that the system does not close the day just because the vehicle stopped. A missing count, route anomaly, late arrival, or handoff issue creates a human-controlled exception. A non-zero cabin count blocks closure until the driver completes the physical rear check and the control room records the outcome.

Four product surfaces, one shared log

The parent app shows trip milestones, absence, collectors, and payments. The crew app supports scanning, checklists, route navigation, and exceptions. The school console handles gate scanning and dismissal. The vehicle unit runs GPS, RFID, camera, IMU, and local safety models. All four surfaces read from the same custody timeline.

6.3 UX and prototype improvements

The parent's home screen should answer one question first: **"Is my child safe right now?"** It should open on a status sentence and the current trip card, not a map that forces the parent to interpret a dot.

The prototype should be localised for India with rupee formatting, Indian names, bilingual labels, school-specific gate flows, and realistic notifications. It should show both successful and failed states.

State to prototype	What the parent should see	Why it matters
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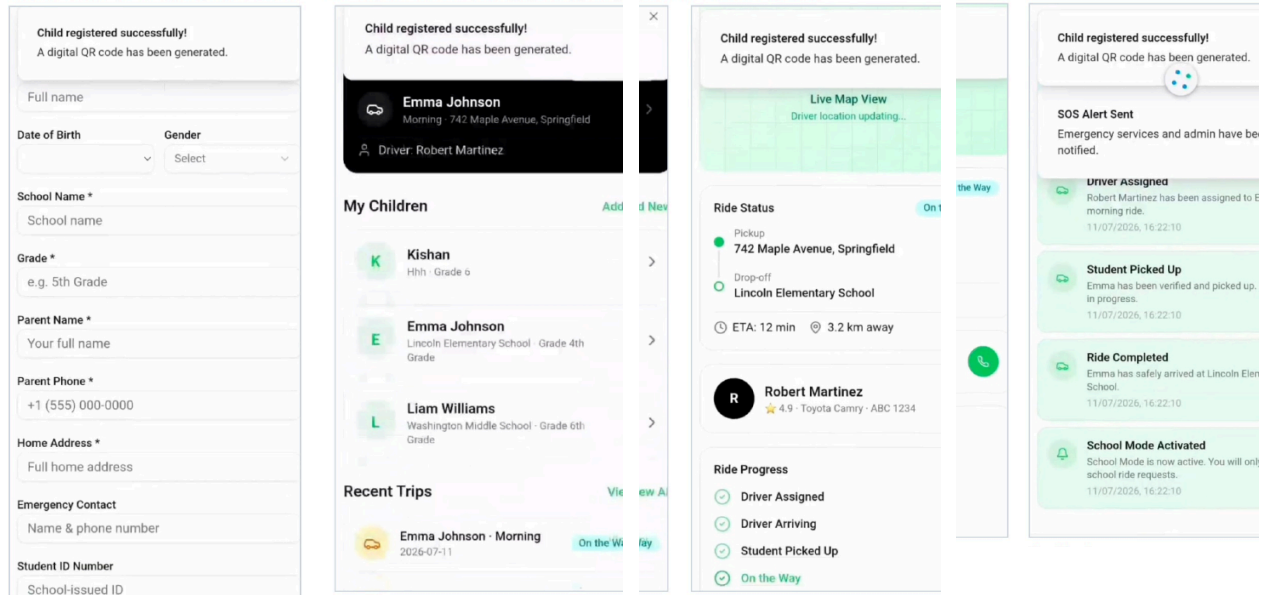
Child absent	Child removed from today's manifest and an absence credit shown.	Prevents false headcount mismatches.
Crew unavailable	Named backup crew, arrival estimate, and human contact.	Makes the recovery promise concrete.
Offline	Last verified event time and clear "manual verification active" status.	Avoids false confidence when live location is unavailable.
Wrong collector	Handoff blocked, reason shown, escalation path visible.	Tests the highest-risk transfer.
Over capacity	Trip blocked with a clear explanation; no silent override.	Makes the capacity USP visible.
Child discomfort	Private escalation route to the attendant/control room.	Tests the pain that a child may never report at home.

The parent-side red SOS button should not be the primary safety mechanism. The child-side silent double-tap band is more relevant, but it should remain P1 until tested with children. The parent app should show headcount status and the daily notification ceiling so the capacity and calm-tracking promises are visible.

6.4 What it looks like: eight interface screens

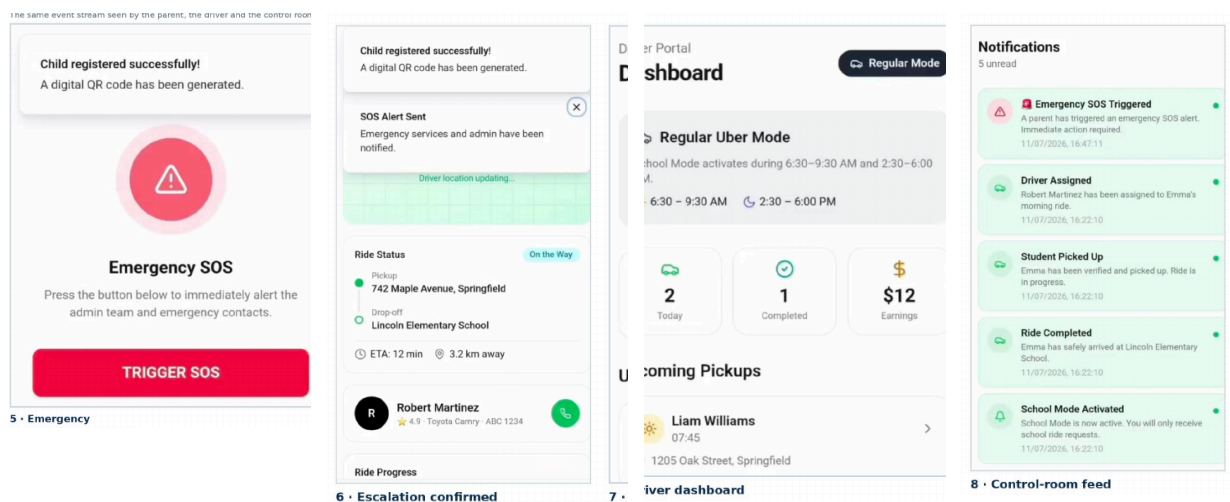
The original prototype is useful because it shows the shared event stream across parent, driver, and control-room surfaces. The screenshots below are retained as reference screens, while the revised product recommendations are described alongside them.

Four screens the parent actually touches. Screens are from the clickable prototype, so copy and placeholder data are not final.



Parent app: registration, household home, live tracking, and alerts.

Screen	What it demonstrates
1 · Register	Child, school, guardian, address, emergency contact, and student ID entry.
2 · Home	Household account, several children, and recent trips in one place.
3 · Track	Live trip status, crew identity, route progress, and ETA.
4 · Alerts	The handoff ledger rendered as a calm event timeline.



Driver and control-room surfaces: emergency, escalation, School Mode, and event feed.

Screen	What it demonstrates
5 · Emergency	Parent-side emergency trigger; the revised product makes the child band the more important signal.
6 · Escalation confirmed	Control-room and emergency-contact acknowledgement with trip context still visible.
7 · Driver dashboard	School Mode, upcoming pickups, earnings, and ordinary-mode suppression during school hours.
8 · Control room	The same event stream monitored by operations and safety staff.

Prototype changes before build Localise the screens for India, replace placeholder US data, make headcount and capacity status visible, show the five-notification ceiling, and add offline, wrong-collector, over-capacity, absent-child, and driver-unavailable states.

Step 7 — Identify KPIs

North star

The north star is **Weekly Verified Safe Handoffs**: a completed journey in which every handoff is within the service-level agreement and no unresolved exception remains.

The metric should be supported by a formal dictionary defining “handoff,” “exception,” “verified,” “late,” “wrong collector,” “manual fallback,” and “child left behind.”

Metric	Initial pilot threshold	Reason
Children left behind	0 absolute	One occurrence triggers a product review and pause.
Verified handoff rate	≥ 99.5%	Measures custody quality rather than only ride completion.
Broken handoff resolved under 5 minutes	≥ 98%	Measures recovery, not just detection.
Over-capacity departures	0 allowed	The vehicle must enforce the promise.

SOS response time	< 90 seconds	The control room must act quickly after a child signal.
False alarms per pod per week	< 1	Too many false alarms destroy trust.
Seat utilisation	≥ 82%	Determines whether the vehicle covers fixed costs.
Contribution margin	≥ 18%	Protects the operating model without trading away safety.

These are internal pilot thresholds, not external industry benchmarks. They should be reviewed after baseline data. Avoid using app sessions and time-in-app as success metrics; a calm parent may use the app less often because the service is working.

Step 8 — Advantages and disadvantages

Sarathi should win on verified adults, accountability, reliability, and parent visibility. It will likely lose initially on price, immediate availability, and route flexibility.

Strength	Why it matters	Risk or trade-off
Locked crew	Familiarity creates trust and a child may speak more easily.	Reduces fleet flexibility and needs a backup crew.
Handoff ledger	Accountability has a visible address.	Requires school and device participation.
Capacity interlock	Overloading becomes structurally difficult.	Adds friction and hardware cost.
On-device AI	Sensitive media stays in the vehicle.	Needs local accuracy testing and maintenance.
School partnership	Adds trust and distribution.	Schools may demand control or revenue share.
Density-first launch	Protects utilisation and contribution.	Growth appears slower at the start.

Main risks

The three highest risks are insufficient density, attendant attrition, and a safety incident. The next risks are parent price resistance and schools deciding to operate the system themselves. The pre-mortem should remain part of the board review, not a one-time slide.

Risk that technology cannot fully solve A child may feel uncomfortable with an adult and never report it. The strongest mitigation is careful hiring, safeguarding training, familiar crews, attentive attendants, and a culture that takes small signals seriously.

Step 9 — The AI feature and privacy

AI should be small, local, assistive, and easy to explain. Sarathi does not need a large autonomous AI system. It needs a few focused helpers that reduce delay, missed communication, and operational uncertainty without making safety decisions alone.

9.1 Which model should Sarathi use?

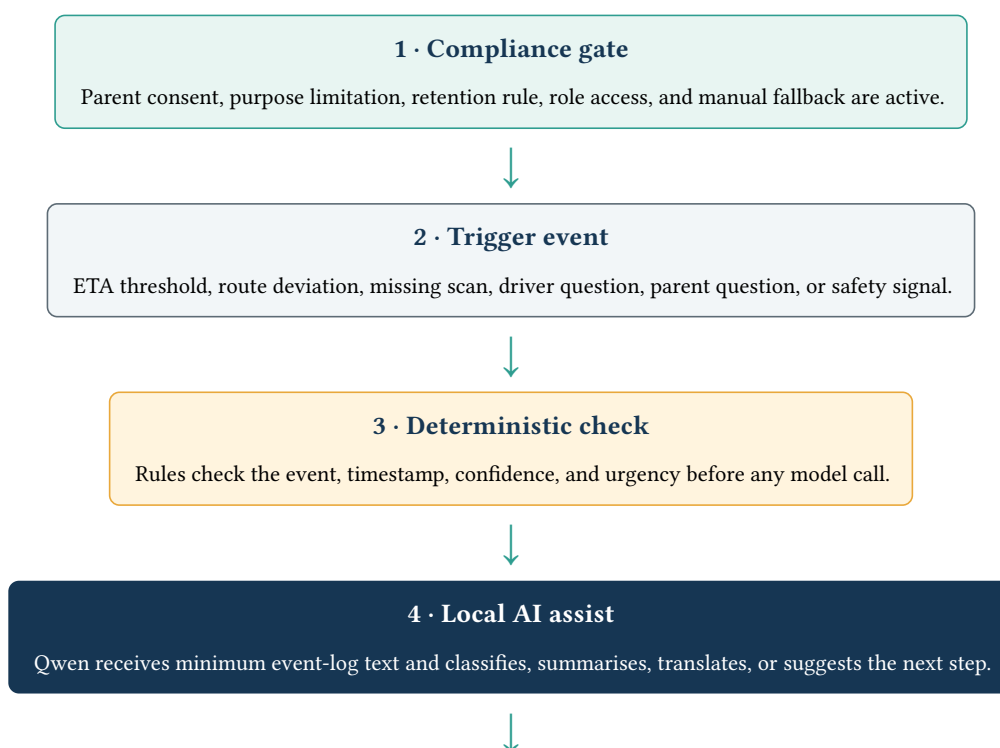
A candidate general-purpose model is **Qwen2.5-3B-Instruct**, **quantised and deployed locally** on a control-room or operations server [model card]. It may support short parent and driver questions, incident summaries, bilingual Hindi-English assistance, and checklist drafting, but these uses require task, language, privacy, and safety evaluation. It should not be trusted for child identity, emotion recognition, or autonomous escalation.

For optional vehicle signals, use a small, task-specific model only after the pilot proves the need. A compact object-detection model such as **YOLOv8n** [official documentation] could support occupancy or belt-check experiments on-device, but it must remain a secondary signal and requires real-vehicle validation. Routing should use deterministic optimisation and traffic data rather than a large language model.

Use case	Recommended approach	Why
Parent and driver questions	Qwen2.5-3B-Instruct, quantised, local	Small enough for local deployment; useful for short, multilingual support answers.
Incident summary	Same local model over event-log text	Summarises what happened without sending cabin media to a third party.
Proactive delay detection	Rules + ETA model + local assistant	A red alert can be triggered from schedule and route deviation; the model explains the reason.
Vehicle occupancy signal	Optional YOLOv8n-style edge model	Task-specific and local; never the only safety control.
Routing	Deterministic route optimisation and traffic data	More predictable and auditable than asking a language model to route a vehicle.

9.2 How the AI system works

The AI does not run before the compliance gates are met. First, parent consent, data purpose, retention limits, role access, and the manual fallback are configured. Then the event stream can trigger a small local model.



5 · Action with human control

Parent gets a red delay alert; driver gets a short prompt; control room approves, escalates, or overrides.



6 · Log and improve

Outcome, source event, timestamp, human decision, and false-alarm result return to the custody log.

AI cycle: from a verified event to a human-controlled action and back to the operating log.

The cycle replaces repetitive event-log reading, repeated ETA messages, first-line question handling, and incident summarisation. It does **not** replace the driver, attendant, safety lead, deterministic hard stops, or the human decision during an emergency.

Situation	AI action	Human-facing result
ETA slips beyond the threshold	Compare scheduled arrival, live route, and corridor status.	Parent receives a red “running late” alert with reason and revised ETA; control room sees the same event.
Route blocked or diverted	Suggest a pre-approved diversion and create a reason code.	Driver receives a short prompt; operations approves if the route changes materially.
Parent asks “Where is my child?”	Read the latest verified event from the custody log.	Answer with the last confirmed milestone, not an invented location.
Driver reports a problem	Summarise the issue and suggest the next checklist step.	Driver gets a clear action; low-confidence cases escalate to a human.
Missing scan or manifest mismatch	Flag the missing event and prevent silent closure.	Crew and control room see the required correction before the trip continues.

The principle is **AI flags and explains; humans decide**. The system should prefer a transparent rule-based red alert over a confident but opaque prediction. Every AI answer should show its source event and timestamp.

9.3 DPDP-aware local deployment

Section 9 of the Digital Personal Data Protection Act, 2023 states that a Data Fiduciary must obtain **verifiable consent of the parent or lawful guardian** before processing a child’s personal data; it also prohibits processing likely to cause detrimental effect on a child’s well-being, and prohibits tracking or behavioural monitoring of children or targeted advertising directed at children, subject to statutory exceptions. Sarathi should therefore use purpose limitation, minimum necessary data, access controls, retention limits, consent withdrawal, and human fallback. This is a product-design interpretation, not legal advice; the final implementation needs qualified legal review. [DPDP Act Section 9] [DPDP Rules 2025]

The proposed architecture would keep child images, audio, and biometric signals inside the vehicle or within Sarathi-controlled infrastructure wherever possible; this is a design intention, not a claim that privacy is automatic. The local Qwen model receives event-log text, not raw cabin media. If a parent withdraws consent or the model is unavailable, the service falls back to manual attendant and control-room procedures.

AI conclusion Sarathi is not building a large AI product. It is using a small locally deployable model, subject to model-license review, for proactive communication and operational assistance, while keeping safety-critical decisions rule-based, human-reviewed, and DPDP-aware.

Step 10 — Summarise

First 30 days

Week 1: speak with three school transport coordinators and test whether the free console creates a real distribution and countersignature path.

Week 2: ride along on six informal routes and interview 20 parents and eight operators to verify the actual load, route, price, and pain points.

Week 3: test all three price tiers with live payment mandates on a waitlist rather than relying only on survey intent.

Week 4: run ten applicants through the complete vetting process and time every stage. This defines the real growth ceiling.

Roadmap

After the safety core is boring and reliable, add pod self-formation, sibling routing across schools, a WhatsApp bot, and broader edge monitoring. Only after density is proven should Sarathi add tuition trips, special-needs mode, bundled insurance, or a B2B “Sarathi for Schools” licence.

Final recommendation

Build Sarathi as a narrow, parent-paid, school-supported route network with a locked crew, hard capacity, verified handoffs, and a human escalation layer. Launch in one dense Gurugram pocket and refuse to expand until safety, density, and false-alarm gates are met.

The product’s strongest USP is not a single feature. It is the combination of **familiar crew, school countersignature, vehicle-enforced capacity, custody ledger, and privacy-aware processing**.

Closing thesis Ride-hailing solved “How do I get there?” For a seven-year-old, the more important question is “Who has my child right now, and who confirmed it?”

Appendix: presentation checklist

Before presenting, be ready to explain the following in your own words: why adult ride-hailing fails for young children; why custody is the core problem; why a locked crew is more valuable than dynamic matching; why the school must participate; why density controls the economics; what the MVP excludes; how the vehicle handles a mismatched count; and which assumptions could kill the business.

References

The following sources were used to check external claims or frame the product methods. Company metrics are treated as company-reported claims; market-size, pricing, capacity, driver-supply, and KPI figures in this assignment remain pilot assumptions unless explicitly sourced.

1. [[Zum official site](#)] — Connected Mobility Experience, school-district model, current public metrics, and 2025 ride count.
2. [[HopSkipDrive official site](#)] — District-led supplemental transportation, CareDrivers, safety positioning, and current public scale claims.
3. [[Digital Personal Data Protection Act, 2023, official text](#)] — Section 9 child-data obligations.
4. [[Digital Personal Data Protection Rules, 2025, official text](#)].
5. [[Qwen2.5-3B-Instruct official model card](#)] and [[Qwen2.5 official announcement](#)].

6. [Ultralytics YOLOv8 documentation] — model-family reference only; no Sarathi performance is implied.
7. [Lewis C. Lin, “Customer Journey Map: The 5Es Framework”] — Entice, Enter, Engage, Exit, Extend.
8. [Lewis C. Lin, The Product Manager Interview, 4th Edition] — product design, market entry, metrics, estimation, and behavioural interview practice.
9. [Lewis C. Lin official books list] — used to verify titles and authorship. It does not list a separate book titled **Product Sense** by Lewis C. Lin; the assignment uses product-sense principles and the 5Es article rather than attributing an unverified book.

Executive summary

Sarathi is a scheduled, parent-paid school-commute service for children in Classes 1–5. Parents pay, children ride, drivers and attendants operate the route, and schools support dismissal and handoff confirmation.

The central problem is not only transportation. It is the parent’s uncertainty about who has the child, whether the route is proceeding normally, and whether the final handoff was completed. Sarathi responds with a familiar crew, fixed corridors, hard capacity, a custody ledger, and human escalation.

The product in one sentence A fixed-route service in which the same crew carries the same pod and every custody transfer is recorded, verified, and recoverable when something goes wrong.

What the pilot proves

The first pilot is intentionally narrow: one dense Gurugram pocket, three partner schools, school trips only, one primary vehicle tier, manual crew vetting, and a staffed control room. The business hypothesis is that route density and repeat trust can make a mixed operator supply base reliable without Sarathi owning the fleet.

Product element	Audited v1 decision
Buyer	Parent or guardian; payment willingness must be tested.
Primary users	Parent, child, driver/attendant; school is a later gatekeeper and trust partner.
Route	Fixed corridor, school commute only.
Safety proof	Manifest reconciliation, verified handoffs, empty-cabin closure, and human escalation.
AI boundary	Local, assistive, event-log based, and never the sole safety decision-maker.
Expansion gate	Do not scale until safety, density, false-alarm, retention, and contribution thresholds are supported by pilot data.

Final assessment

The strongest USP is not a single app screen. It is the operating combination of familiar crews, school countersignatures, vehicle capacity, a custody ledger, proactive communication, and privacy-aware processing. The largest remaining uncertainty is not whether the concept is understandable; it is whether parents will pay, schools will participate, and mixed vehicle operators can deliver the promised consistency at route-level density.

All market sizes, price bands, capacity tiers, driver-supply ranges, and KPI thresholds in this assignment are planning assumptions or pilot targets unless a source is explicitly cited. The document should be updated after field interviews, ride-alongs, payment tests, and a controlled route pilot.